D3 components create elements in the DOM and how they're classed so that you can style them properly.

4.3 Complex graphical objects

Using circles or rectangles for your data won't work with some datasets, for example, if an important aspect of your data has to do with distribution, like user demographics or statistical data. Often, the distribution of data gets lost in information visualization, or is only noted with a reference to standard deviation or other first-year statistics terms that indicate the average doesn't tell the whole story. One particularly useful way of representing data that has a distribution (such as a fluctuating stock price) is the use of a boxplot in place of a traditional scatterplot. The boxplot uses a complex graphic that encodes distribution in its shape. The box in a boxplot typically looks like the one shown in figure 4.11. It uses quartiles that have been preprocessed, but you could easily use `d3.scale.quartile()` to create your own values from your own dataset.

Take a moment to examine the amount of data that's encoded in the graphic in figure 4.11. The median value is represented as a gray line. The rectangle shows the amount of whatever you're measuring that falls in a set range that represents the majority of the data. The two lines above and below the rectangle indicate the minimum and maximum values. Everything except the information in the gray line is lost when you map only the average or median value at a datapoint.

To build a reasonable boxplot, we'll need a set of data with interesting variation in those areas. Let's assume we want to plot the number of registered visitors coming to our website by day of the week so that we can compare our stats week to week (or so that we can present this info to our boss, or for some other reason). We have the data

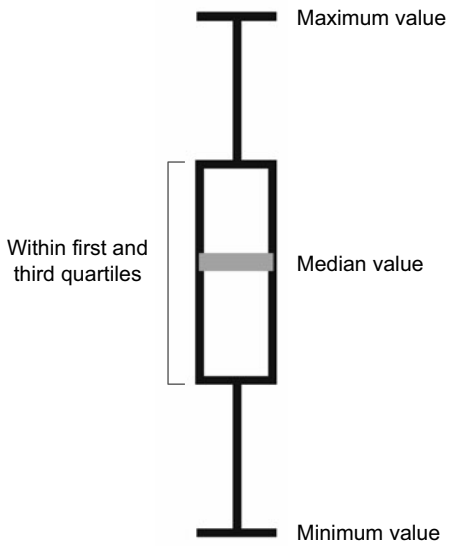


Figure 4.11 A box from a boxplot consists of five pieces of information encoded in a single shape: (1) the maximum value, (2) the high value of some distribution, such as the third quartile, (3) the median or mean value, (4) the corresponding low value of the distribution, such as the first quartile, and (5) the minimum value.